# VISVESVARAYA TECHNOLOGICAL UNIVERSITY

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Lab Experiment Record

## Project Management with Git [BCSL58C]

*Submitted in partial fulfillment towards AEC of 3rd semester of*

**Bachelor of Engineering**
**in**
**Computer Science and Engineering**
**(Artificial Intelligence & Machine Learning)**

Submitted by


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**(Accredited by NAAC)**


**2025-2026**

**Program 1: Setting Up and Basic Commands**

Create a new folder in a directory. Open GitBash from there.

Run the init command:

git init

Give your identity ie., your username and your email ID.

git config user.email  “bhavwork897@gmail.com”

git config user.name "bhavwork"

**Procedure**

1. Create a new folder in the desired directory and open **Git Bash** inside that folder.
2. Initialize a new Git repository by executing the command git init.
3. Configure the user identity by setting the username and email ID using git config user.name and git config user.email.
4. Check the current status of the repository using the git status command.
5. Add the required file(s) to the staging area using the git add <file-name> command.
6. Commit the staged changes to the repository with a meaningful commit message using git commit -m "Initial commit".

git init

git status

git add <file-name>

git commit -m "Initial commit"

This experiment initializes a new Git repository, adds a file to the staging area, and commits the changes with an appropriate commit message.

```
mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gitlab
$ git config --global user.name "bhavwork"

mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gitlab
$ git config --global user.email "bhavwork897@gmail.com"

mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gitlab
$ git remote add origin "https://github.com/bhavwork/gitpractice.git"
fatal: not a git repository (or any of the parent directories): .git

mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gitlab
$ git init
Initialized empty Git repository in D:/Bhavana PSNL/gitlab/.git/

mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gitlab (master)
$ git remote add origin "https://github.com/bhavwork/gitpractice.git"

mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gitlab (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        hello.txt

nothing added to commit but untracked files present (use "git add" to track)

mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gitlab (master)
$ git add .

mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gitlab (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   hello.txt


mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gitlab (master)
$ git commit -m "added hello txt file"
[master (root-commit) ae485f6] added hello txt file
 1 file changed, 0 insertions(+), 0 deletions(-)
```

## Program 2: Creating and Managing Branches

**Procedure:**

git branch feature-branch

git checkout master

git merge feature-branch

```
$ git branch
* master

mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gitlab (master)
$ git branch feature

mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gitlab (master)
$ git branch
  feature
* master

mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gitlab (master)
$ git checkout feature
Switched to branch 'feature'

mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gitlab (feature)
$ git add words
fatal: pathspec 'words' did not match any files

mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gitlab (feature)
$ git add words.txt

mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gitlab (feature)
$ git commit -m "uploaded words in feature branch"
[feature e72907a] uploaded words in feature branch
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 words.txt

mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gitlab (feature)
$ git checkout main
error: pathspec 'main' did not match any file(s) known to git

mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gitlab (feature)
$ git checkout master
Switched to branch 'master'

mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gitlab (master)
$ git merge feature
Updating ae485f6..e72907a
Fast-forward
 words.txt | 0
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 words.txt

mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gitlab (master)
$ git status
On branch master
nothing to commit, working tree clean
```

git branch feature-branch

git checkout feature-branch

git checkout main

git merge feature-branch

This experiment demonstrates how to create a new branch, switch between branches, and merge a feature branch into the main branch.

**Program 3: Stashing Changes**

git stash

git checkout feature-branch

git stash apply

```
mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gitlab (master)
$ git stash
No local changes to save

mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gitlab (master)
$ git stash
Saved working directory and index state WIP on master: e72907a uploaded words in feature branch

mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gitlab (master)
$ git status
On branch master
nothing to commit, working tree clean

mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gitlab (master)
$ git checkout feature
Switched to branch 'feature'

mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gitlab (feature)
$ git stash apply
On branch feature
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   hello.txt
        modified:   words.txt

no changes added to commit (use "git add" and/or "git commit -a")
```

This experiment temporarily stores uncommitted changes using Git stash, switches branches safely, and restores the stashed changes.

**Program 4: Cloning a Remote Repository**

git clone <repository-url>

```
mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gitlab (feature)
$ git clone "https://github.com/SnehagangaNS/project-demo.git"
Cloning into 'project-demo'...
remote: Enumerating objects: 14, done.
remote: Counting objects: 100% (14/14), done.
remote: Compressing objects: 100% (13/13), done.
remote: Total 14 (delta 1), reused 13 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (14/14), 5.07 KiB | 1.27 MiB/s, done.
Resolving deltas: 100% (1/1), done.
```

This experiment clones a remote Git repository to the local machine, including all files, branches, and commit history.

**Procedure**

1. Open **Git Bash** in the required directory.
2. Copy the URL of the remote Git repository to be cloned.
3. Execute the command git clone <repository-url> to create a local copy of the remote repository.
4. Navigate into the cloned repository using the cd <repository-name> command.
5. Verify the successful cloning by checking the repository contents or using git status.

**Program 5: Fetch and Rebase**

git fetch origin

git rebase origin/main

This experiment fetches the latest changes from the remote repository and rebases the local branch onto the updated remote branch to maintain a clean history.

```
mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gsss (feature)
$ git status
On branch feature
nothing to commit, working tree clean

mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gsss (feature)
$ git fetch origin

mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gsss (feature)
$ git rebase origin/main
Current branch feature is up to date.

mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gsss (feature)
$ git stash
No local changes to save

mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gsss (feature)
$ git rebase origin/main
Current branch feature is up to date.

mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gsss (feature)
$ git status
On branch feature
nothing to commit, working tree clean
```

**Program 6: Merge with Custom Message**

git checkout main

git merge feature-branch -m "Merge feature-branch into main"

```
mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gsss (feature)
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gsss (main)
$ git merge feature -m "merged feature to main"
Updating f01311c..696b3df
Fast-forward (no commit created; -m option ignored)
 words.txt | 0
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 words.txt

mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gsss (main)
$ git log --oneline --graph
* 696b3df (HEAD -> main, feature) added words in feature
 branch
* f01311c (origin/main) added hello text file
```

This experiment merges a feature branch into the main branch while providing a custom merge commit message

**Program 7: Git Tags**

Aim:

To understand and demonstrate the use of Git tags for marking specific points in a repository's commit history, such as important versions or releases.

Command:

git tag v1.0

```
mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gsss (main)
$ git log --oneline
696b3df (HEAD -> main, feature) added words in feature
ranch
f01311c (origin/main) added hello text file

mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gsss (main)
$ git tag v1.0

mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gsss (main)
$ git tag
v1.0

mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gsss (main)
$ |
```

This experiment creates a lightweight Git tag named v1.0 that points to a specific commit in the repository.

**Program 8: Cherry-pick Commits**

git cherry-pick <start-commit>^..<end-commit>

```
mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gsss (main)
$ git log feature --oneline
696b3df (HEAD -> main, tag: v1.0, feature) added words i
n feature branch
f01311c (origin/main) added hello text file

mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gsss (main)
$ git log main --oneline
696b3df (HEAD -> main, tag: v1.0, feature) added words i
n feature branch
f01311c (origin/main) added hello text file

mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gsss (main)
$ git cherry icl 696b3df
fatal: unknown commit icl

mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gsss (main)
$ git cherry-pick 696b3df
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

You are currently cherry-picking commit 696b3df.
  (all conflicts fixed: run "git cherry-pick --continue"
)
  (use "git cherry-pick --skip" to skip this patch)
  (use "git cherry-pick --abort" to cancel the cherry-pi
ck operation)

nothing to commit, working tree clean
The previous cherry-pick is now empty, possibly due to c
onflict resolution.
If you wish to commit it anyway, use:

    git commit --allow-empty

Otherwise, please use 'git cherry-pick --skip'
```

This experiment applies a selected range of commits from a source branch onto the current branch using cherry-pick.

**Program 9: View Commit Details**

git show <commit id>

```
mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gsss (main|CHERR
Y-PICKING)
$ git show 696b3df
commit 696b3df158fa2d6016033a4eec2760e519f22eba (HEAD -> main,
 tag: v1.0, feature)
Author: bhavwork <bhavwork897@gmail.com>
Date:   Tue Jan 6 10:50:09 2026 +0530

    added words in feature branch

diff --git a/words.txt b/words.txt
new file mode 100644
index 0000000..e69de29

mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gsss (main|CHERRY-PICK
```

This shows the details made by that commit

**Program 10: List Commits by Author**

git log --author="JohnDoe" --since="2023-01-01" --until="2023-12-31"

```
mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gsss (main|CHERRY-PICK
ING)
$ git log --author="JohnDoe" --since="2023-01-02" --until="202
3-12-31" --oneline
```

This experiment displays all commits made by a specific author within a given date range.

## Program 11: Last Five Commits

git log -5

```
mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gsss (main|CHERRY-PICK
ING)
$ git log -5
commit 696b3df158fa2d6016033a4eec2760e519f22eb
a (HEAD -> main, tag: v1.0, feature)
Author: bhavwork <bhavwork897@gmail.com>
Date:   Tue Jan 6 10:50:09 2026 +0530

    added words in feature branch

commit f01311c156d5cb1955984045367d5a73c19a9b0
0 (origin/main)
Author: bhavwork <bhavwork897@gmail.com>
Date:   Tue Jan 6 10:45:32 2026 +0530

    added hello text file

mprak@RAKSHITHA MINGW64 /d/Bhavana PSNL/gsss (
main|CHERRY-PICKING)
$
```

This experiment displays the most recent five commits in the repository's history

## Program 12: Undo Commit

git revert abc123

This experiment undoes the changes introduced by a specific commit by creating a new commit that reverses those changes.

```
[main cecdd89] Revert "Updated this text fild in deature-branch"
 1 file changed, 1 deletion(-)
```